GRADUATION PROJECT · TEAM PLAN

Bike Sales Analysis & Business Intelligence Dashboard

Team Roles and Project Execution Plan

Power BI · Power Query · SQL Server · Excel · Tableau (optional)

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1. Project Planning & Management

1.1 Project Title

Bike Sales Analysis & Business Intelligence Dashboard.

1.2 Project Purpose

The project turns raw bike sales records into a clear, interactive Power BI report focused on SALES ANALYSIS. It helps a business audience understand sales value and quantity over time, by bike model and store, and across the available customer, salesperson and payment dimensions. The team will complete the analytics workflow: data cleaning, modeling, validation, analysis, visualization and documentation.

1.3 Main Objectives

- Identify sales trends over time (year, quarter, month, day of week).
- Measure product (bike model) and store performance and highlight top and low performers.
- Understand customers through the demographic fields that exist in the data.
- Evaluate salesperson performance using the salesperson identifiers in the data.
- Examine payment method patterns and their relation to sales.
- Deliver one consistent, validated and well-documented Power BI report.

1.4 Tools and Technologies

Tool	Role in the project
Power Query (Power BI)	Data import, cleaning and transformation.
Power BI	Data model, DAX measures, KPI cards, charts, dashboard design and the final report.
SQL Server	Queries for validation, aggregation and analysis.
Excel	PivotTables, supporting charts and cross-checks of key figures.
Tableau (optional)	Complementary visualization or comparison, used only if it adds clear value.
GitHub	Version control and the shared project repository.

1.5 Dataset Snapshot

The team received a seven-table star schema (BikeSales_7Table_Model_1) derived from a source file of 100,000 sales rows. The facts below were read directly from the supplied files and should be re-confirmed by Member 1 and Member 2 during Phase 1.

Table	What the supplied files show
FactSales	100,000 rows. Fields: Sale_ID, DateKey, CustomerKey, BikeKey, StoreKey, SalespersonKey, PaymentKey, UnitPrice, Quantity (1 to 5), TotalSales (UnitPrice × Quantity).
DimDate	Daily calendar, 1 Jan 2020 to 22 Sep 2024. Fields: Date, Year, Quarter, MonthNumber, MonthName, YearMonth, Day, DayOfWeekNumber, DayName, IsWeekend.
DimBike	7 bike models: BMX, Cruiser, Electric Bike, Folding Bike, Hybrid Bike, Mountain Bike, Road Bike.
DimStore	7 store locations: Chicago, Houston, Los Angeles, New York, Philadelphia, Phoenix, San Antonio.
DimPayment	6 payment methods: Apple Pay, Cash, Credit Card, Debit Card, Google Pay, PayPal.
DimSalesperson	900 salespeople, identified by Salesperson_ID only (no name field in the supplied table).
DimCustomer	94,960 rows covering 9,000 distinct Customer_ID values. Fields: Customer_Age (18 to 70), Customer_Gender (Female, Male), ObservedProfileVariants. Customer_ID is not unique here; CustomerKey is.

> *Important limits of the data: there are no cost, profit, discount, region-of-customer or product-category fields in the supplied tables, so metrics such as profit margin or discount impact must not be reported. Customer analysis must count Customer_ID (not CustomerKey) to avoid double-counting.*

1.6 Expected Final Deliverable

A single integrated Power BI report with four analytical pages (Sales Overview, Product & Store Performance, Customer & Salesperson Insights, Payment Analysis), supported by documented Power Query steps, SQL scripts and validation notes, Excel summaries, a data-quality report, project documentation and an organized GitHub repository. An optional Tableau comparison may be added if justified.

1.7 Project Scope

Scope: SALES ANALYSIS using the supplied bike transactions. Deliver a validated, interactive Power BI report for sales trends, bike models/stores, available customer and salesperson data, and payment methods. Profit/margin, discounts, inventory, returns, customer location, forecasting and market share are excluded unless source fields are added.

1.8 Execution Timeline (19 October - 1 December 2026)

Dates	Phase	Lead / contributors	Milestone / acceptance output
19-23 Oct	Plan & scope lock	Member 1 + team	Confirm scope, requirements and repo.
24-30 Oct	Clean & model	Members 1-2	Log cleaning; validate keys and row count.
31 Oct-6 Nov	KPI validation	Members 2-3; 4-6 review	Reconcile core KPIs in SQL/Excel.
7-13 Nov	Build report pages	Members 3-6	Build four pages with shared measures/filters.
14-20 Nov	Integrate & test	Member 6 + team	Test filters, navigation and figures.
21-27 Nov	Fix & document	Members 1-2 + team	Close issues; finalize scripts and notes.
28 Nov-1 Dec	Final QA & release	Member 1 coordinates	Approve PDF, GitHub package and walkthrough.

1.9 Key Risks and Mitigation

Risk	Likelihood / impact	Mitigation / response
Repeated Customer_ID profiles	Medium / High	Count distinct Customer_ID; document treatment of age/gender variants.
Power BI / SQL / Excel totals differ	Medium / High	Agree KPI definitions and reconcile totals before sign-off.
Analysis or page delivery is late	Medium / High	Track milestones weekly and raise blockers early.
Scope creep beyond sales analysis	Medium / Medium	Include only metrics supported by confirmed source fields.
GitHub conflicts or exposed credentials	Low-Med / High	Use one repo and regular commits; never commit passwords/tokens.

1.10 Project Success Criteria

- All four report pages are integrated; navigation and slicers work.
- Core sales KPIs reconcile to SQL/Excel; remaining differences are documented.
- Fact-row baseline and relationships are validated; data limits are disclosed.
- Cleaning/model notes, measures, validation scripts and three documentation parts are in GitHub.

2. Team Structure and Role Distribution

Responsibilities are divided among exactly six members. Each member owns a defined part of the work and produces a visible deliverable. Cross-member dependencies and collaboration partners are summarized in the Responsibility Matrix (Section 3).

Team Roster

Member	Name	Role
Team Leader (Member 1)	بسمله احمد محمود محمد	Team Leader & Data Cleaning Specialist
Member 2	ايمان عماد منصور عبدالحليم	SQL & Data Modeling Specialist
Member 3	نسرين سمير احمد ابوهديمه	Sales Analysis Specialist
Member 4	ضحى ياسر فتح الله سيد	Product & Store Performance Analyst
Member 5	ندى مجدى أبوقريش بيومى	Customer & Salesperson Analyst
Member 6	نور الهدي ياسر زكي أحمد	Payment Analysis & Dashboard Integration Specialist

Member 1: Team Leader & Data Cleaning Specialist

Team member: بسمله احمد محمود محمد

Main tools: Power Query in Power BI, Excel, GitHub.

Responsibilities

- Coordinate team tasks, deadlines, progress and repository organization.
- Inspect the dataset and clean data types, missing values, duplicates, inconsistent formats and other quality issues in Power Query.
- Document cleaning decisions and prepare clean data for analysis.
- Track integration and delivery status without taking over other members' analytical work.

Expected deliverables

- Cleaned dataset / documented Power Query steps and data-quality report.
- Team task tracker and organized GitHub repository.

Member 2: SQL & Data Modeling Specialist

Team member: ايمان عماد منصور عبدالحليم

Main tools: SQL Server and Power BI.

Responsibilities

- Review table structures, primary/foreign keys and relationships; connect data in SQL Server where appropriate.
- Write SQL for validation, aggregation, filtering and analysis; reconcile key figures across SQL, Power BI and Excel.
- Document SQL queries, data model and validation results.

Expected deliverables

- SQL scripts and validation results.
- Data model and relationship documentation.

Member 3: Sales Analysis Specialist

Team member: نسرين سمير احمد ابوهديمه

Main tools: Power BI, SQL Server and Excel.

Responsibilities

- Analyze total sales, quantity sold and trends by date, month and year.
- Define supported KPIs (Total Sales, Quantity Sold and Average Transaction Value); validate figures in Excel/SQL.
- Build the Sales Overview Power BI page and supporting summaries.

Expected deliverables

- KPI calculations and Excel summaries.
- Sales Overview dashboard page.

Member 4: Product & Store Performance Analyst

Team member: ضحى ياسر فتح الله سيد

Main tools: Power BI, SQL Server and Excel.

Responsibilities

- Compare bike models and stores by sales and quantity; identify top/low performers when supported.
- Use Excel summaries and validate rankings/totals with SQL.
- Build the Product & Store Performance Power BI page.

Expected deliverables

- Product/store analysis and supporting summaries.
- Product & Store Performance dashboard page.

Member 5: Customer & Salesperson Analyst

Team member: ندى مجدى أبوقريش بيومى

Main tools: Power BI, SQL Server and Excel.

Responsibilities

- Analyze customer age/gender fields and salesperson performance using Salesperson_ID.
- Use distinct Customer_ID for unique customer counts; validate counts and calculations with SQL/Excel.
- Build the Customer & Salesperson Insights Power BI page.

Expected deliverables

- Customer/salesperson analysis and Excel summaries.
- Customer & Salesperson Insights dashboard page.

Member 6: Payment Analysis & Dashboard Integration Specialist

Team member: نور الهدي ياسر زكي أحمد

Main tools: Power BI, SQL Server and optionally Tableau.

Responsibilities

- Analyze sales and transactions by payment method and build the Payment Analysis page.
- Integrate all pages; maintain a consistent design and review navigation, readability and usability.
- Use Tableau only if a complementary comparison adds value; Tableau is optional.

Expected deliverables

- Payment Analysis page and integrated Power BI report.
- Optional Tableau analysis, if justified.

3. Responsibility Matrix

The matrix summarizes each member's role at a glance. Effort is spread across data preparation, modeling, four analytical pages and integration.

Team Member	Main Role	Primary Tools	Main Responsibilities	Expected Deliverables	Collaborates With
Member 1 **بسمله احمد محمود محمد**	Team Leader & Data Cleaning Specialist	Power Query, Excel, GitHub	• Coordinate tasks and deadlines • Import, inspect and clean data • Document cleaning decisions • Organize files in GitHub	• Cleaned data / Power Query steps • Data quality report • Task tracker and repository	Members 2, 3, 4, 5, 6
Member 2 **ايمان عماد منصور عبدالحليم**	SQL & Data Modeling Specialist	SQL Server, Power BI	• Review tables, keys, relationships • Write validation and aggregation queries • Cross-check key figures	• SQL scripts • Data model documentation • Validation results	Members 1, 3, 4, 5, 6
Member 3 **نسرين سمير احمد ابوهديمه**	Sales Analysis Specialist	Power BI, SQL Server, Excel	• Sales and quantity analysis • Time trends and KPIs • Excel verification	• KPI calculations • Excel summaries • Sales Overview page	Members 2, 4, 5, 6
Member 4 **ضحى ياسر فتح الله سيد**	Product & Store Performance Analyst	Power BI, SQL Server, Excel	• Bike model analysis • Store comparison • Top and low performers	• Product and store analysis • Excel summaries • Product & Store Performance page	Members 2, 3, 6
Member 5 **ندى مجدى أبوقريش بيومى**	Customer & Salesperson Analyst	Power BI, SQL Server, Excel	• Customer demographics • Salesperson performance • Correct customer counting	• Customer and salesperson analysis • Excel summaries • Customer & Salesperson Insights page	Members 1, 2, 3, 6
Member 6 **نور الهدي ياسر زكي أحمد**	Payment Analysis & Dashboard Integration Specialist	Power BI, SQL Server, Tableau (optional)	• Payment analysis • Visual consistency • Report integration and usability review • Optional Tableau	• Payment Analysis page • Integrated Power BI report • Optional Tableau analysis	Members 1, 2, 3, 4, 5

4. Project Execution Workflow

Five phases, each with a clear output the next depends on.

Phase	Activities	Responsible Members	Tools	Outputs
Phase 1: Data Preparation	• Member 1 cleans the data in Power Query • Member 2 reviews table structures, keys and relationships • Both document data issues and coordinate before analysis begins	Members 1 and 2	Power Query, SQL Server, Excel	• Cleaned data • Data quality report • Relationship documentation
Phase 2: Data Validation and Analysis	• Member 2 prepares SQL queries for validation and aggregation • Members 3, 4, 5 and 6 begin their analyses • Excel is used for supporting summaries and cross-checks • Power BI is used for analytical visuals and KPI calculations	Member 2 (queries); Members 3, 4, 5, 6 (analysis)	SQL Server, Excel, Power BI	• SQL scripts • Draft measures and visuals • Excel cross-check summaries
Phase 3: Dashboard Development	• Each responsible member builds the assigned page • Team uses consistent metric definitions • Results are validated before entering the final report	Members 3, 4, 5, 6	Power BI, SQL Server, Excel	Four validated dashboard pages
Phase 4: Integration and Quality Review	• Member 6 coordinates integration and visual consistency • Member 1 tracks completion and manages project files • All members check for incorrect figures, duplicate counting, missing labels, confusing visuals and inconsistent formatting	Member 6 leads; Member 1 tracks; all members review	Power BI, GitHub	• Integrated Power BI report • Review log of issues fixed
Phase 5: Documentation and Final Submission	• Collect the Power BI report, SQL scripts, Excel files, cleaning documentation, analysis summaries and project documentation • Organize files in a clear folder structure • Upload approved files to the team GitHub repository	All members; Member 1 coordinates	GitHub, Word, Power BI, Excel	• Final submission package • Organized GitHub repository

5. Suggested Power BI Dashboard Structure

The four pages below are suggestions. Every metric and visual depends on the fields actually present in the cleaned data. Field names listed come from the supplied model and must be confirmed before use.

5.1 Sales Overview (Member 3)

Purpose: Give a high-level view of sales volume, value and how they change over time.

Element	Suggestion
Suggested KPIs	• Total Sales (sum of TotalSales) • Quantity Sold (sum of Quantity) • Number of transactions (count of Sale_ID) • Average Transaction Value (Total Sales ÷ transactions) • Year-over-year change, if the full years are comparable (2024 data ends in September)
Suggested charts and visuals	• KPI cards • Line chart of sales by YearMonth • Column chart by Year or Quarter • Day-of-week chart (DayName / IsWeekend) • Slicers for Year and Quarter
Fields to check in the dataset	• FactSales: TotalSales, Quantity, UnitPrice, Sale_ID • DimDate: Year, Quarter, MonthName, YearMonth, DayName, IsWeekend

5.2 Product & Store Performance (Member 4)

Purpose: Show which bike models and stores perform best and worst.

Element	Suggestion
Suggested KPIs	• Sales and quantity by bike model • Sales and quantity by store • Share of total sales per model / store • Average unit price by model
Suggested charts and visuals	• Ranked bar charts (model, store) • Matrix of store by bike model • Treemap or donut for share of sales • Trend of top models over time • Slicers for Store and Bike_Model
Fields to check in the dataset	• DimBike: Bike_Model • DimStore: Store_Location • FactSales: TotalSales, Quantity, UnitPrice • Note: no product category, cost or profit fields were found in the supplied tables

5.3 Customer & Salesperson Insights (Member 5)

Purpose: Describe who buys and how sales are distributed across salespeople.

Element	Suggestion
Suggested KPIs	• Distinct customers (distinct count of Customer_ID, not CustomerKey) • Sales by customer gender and age group • Sales per salesperson • Average sales per salesperson • Top and bottom salespeople by Total Sales
Suggested charts and visuals	• Bar charts by gender and age band (age bands created in Power Query or DAX) • Top N / bottom N salesperson chart • Distribution chart of sales per salesperson • Detail table with sorting
Fields to check in the dataset	• DimCustomer: Customer_ID, Customer_Age, Customer_Gender, ObservedProfileVariants • DimSalesperson: Salesperson_ID (no name field supplied) • Check how repeated Customer_ID profiles are handled before reporting any demographic split

5.4 Payment Analysis (Member 6)

Purpose: Show how customers pay and how payment methods relate to sales.

Element	Suggestion
Suggested KPIs	• Sales and transactions by payment method • Share of transactions per method • Average transaction value by method • Payment method mix over time
Suggested charts and visuals	• Donut or bar chart of payment share • Stacked column of payment method by Year • Matrix of payment method by store • Slicers for Year and Store
Fields to check in the dataset	• DimPayment: Payment_Method • FactSales: TotalSales, Sale_ID, PaymentKey • DimDate: Year, YearMonth

> *Do not add any metric whose required field is missing (for example profit, margin, discounts, customer location or product category). If the team wants such a metric, it must first be confirmed in the source data.*

6. Tools and Their Purpose

Tool	How it will be used	Main users
Power Query	Data cleaning and transformation: fix data types, handle missing values and duplicates, standardize formats, and record each applied step.	Member 1 (others review)
Power BI	Data modeling (relationships between fact and dimension tables), DAX measures where needed, KPI cards, charts, dashboard design and final reporting.	Members 3, 4, 5, 6 (model: 2)
SQL Server	Queries, aggregations, validation and data analysis. Simple SELECT, JOIN, GROUP BY and COUNT DISTINCT are sufficient.	Member 2 (others validate)
Excel	Preliminary analysis, PivotTables, supporting charts and cross-checking calculations.	Members 3, 4, 5
Tableau (optional)	Supplementary visualization or comparison, only if it adds clear value to the report.	Member 6
GitHub	Shared repository for scripts, files and documentation.	Member 1 (all contribute)

7. Collaboration and Quality Control Rules

- Each member delivers and explains their work; the leader coordinates and reviews without replacing others' contributions.
- Agree KPI definitions, validate key figures against SQL/Excel, and document unexplained differences.
- Confirm fields before use; record cleaning assumptions and count unique customers with Customer_ID, not CustomerKey.
- Keep one organized GitHub repository with clear filenames and regular commits. Never commit credentials; Tableau is optional.

8. Final Deliverables Checklist

Tick each item when it has been completed, reviewed and uploaded to the repository.

Done	Deliverable	What to confirm	Owner
☐	Data preparation & quality	Cleaned data, documented Power Query steps and quality report	Member 1
☐	SQL, data model & validation	Commented SQL scripts; keys, relationships and validation results documented	Member 2
☐	Excel supporting analyses	PivotTables, summaries and cross-checks for assigned analyses	Members 3-5
☐	Sales Overview page	KPIs verified; labels and filters clear	Member 3
☐	Product & Store Performance page	Rankings verified; filters work	Member 4
☐	Customer & Salesperson Insights page	Distinct Customer_ID counts; results validated	Member 5
☐	Payment Analysis page	Payment figures validated	Member 6
☐	Integrated report & optional Tableau	Consistent design/navigation; Tableau only if justified	Member 6
☐	Project documentation & GitHub	Three documentation parts; organized repo and approved files uploaded	All members / Member 1

Suggested Repository Folder Structure: Keep files under /data, /power-query, /sql, /excel, /powerbi, /tableau (optional), and /docs. Store source/cleaned data or links, transformations, SQL/validation files, Excel analyses, final .pbix, optional Tableau workbook, and documentation in the matching folders.

9. Literature Review

The following official Microsoft Power BI examples were reviewed as comparable sales/retail reporting solutions. They are used as design and analysis references, not as datasets to be merged into this project.

Comparable solution	Strengths / relevance	Limitations for Bike Sales	Application to this project
Sales and Marketing sample [1]	Uses KPI cards, time trends, and breakdowns by region, product/segment and market measures. Its multi-page story is a useful model for executive sales reporting.	Market-share, sentiment and competitor measures are not present in the supplied bike-sales model.	Reuse the KPI + trend narrative and category/store breakdown approach; exclude unsupported market-share or sentiment claims.
Retail Analysis sample [2]	Compares sales and units across stores/districts and includes year-over-year and store-performance views. This closely supports store and sales trend analysis.	Gross margin, variance and new-store analyses require fields or context not confirmed in our dataset.	Adapt store comparisons and sales/quantity trends; do not report margin or new-store effects without matching fields.
Sales and Returns sample [3]	Demonstrates report storytelling, interactivity, tooltips, and a focused sales page structure.	Returns-related measures and other specialized fields are not confirmed in our source data.	Borrow interaction and presentation ideas only; keep the project limited to the sales measures and dimensions available.

Design principle: the supplied model is already organized as one sales fact table with related dimensions. Microsoft's star-schema guidance explains that dimension tables support filtering/grouping while fact tables support summarization [4]. The team will validate the actual relationships and use consistent KPI definitions before building the report.

Research gap and project fit

The reviewed examples show the value of multi-page, interactive sales reporting, but their metrics cannot be copied wholesale. This project focuses on bike sales transactions and the seven supplied tables: time, bike model, store, customer, salesperson and payment. Its contribution is a bike-specific, validated report built only from supported fields; it will not claim profit, discounts, inventory, returns or market-share insights when those data are unavailable.

References

[1] Microsoft Learn. "Sales and Marketing sample for Power BI: Take a tour." https://learn.microsoft.com/en-us/power-bi/create-reports/sample-sales-and-marketing (accessed 19 October 2026).

[2] Microsoft Learn. "Retail Analysis sample for Power BI: Take a tour." https://learn.microsoft.com/en-us/power-bi/create-reports/sample-retail-analysis (accessed 19 October 2026).

[3] Microsoft Learn. "Explore the Sales and Returns sample report in Power BI." https://learn.microsoft.com/en-us/power-bi/create-reports/sample-sales-returns (accessed 19 October 2026).

[4] Microsoft Learn. "Understand star schema and the importance for Power BI." https://learn.microsoft.com/power-bi/guidance/star-schema (accessed 19 October 2026).

10. Requirements Gathering

This section translates the sales-analysis scope into needs, report functions and quality requirements. Requirements are based on the available seven-table bike-sales model and should be confirmed during validation.

10.1 Stakeholders and needs

Stakeholder	Main information need
Sales manager / business decision-maker	Review total sales, trends, and top/low-performing models and stores.
Store / sales supervisor	Compare sales and quantity by store, period, salesperson ID, and payment method.
Sales / BI analyst	Filter reports, inspect trends, and use consistent KPIs and supported dimensions.
Project team / reviewer	Verify data quality and calculations; locate cleaning steps, SQL and documentation in GitHub.

10.2 Functional requirements

ID / requirement	Expected behavior
FR-01 Sales KPIs	Show Total Sales, Quantity Sold, transaction count and Average Transaction Value; document definitions.
FR-02 Time analysis	Show sales/quantity by year, quarter, month and weekday; data ends 22 Sep 2024.
FR-03 Product and store	Rank bike models and stores by sales and quantity, with sales-share views.
FR-04 Customer and salesperson	Show supported age/gender summaries and sales by Salesperson_ID; distinct-count Customer_ID.
FR-05 Payment analysis	Compare sales, transaction count and average transaction value by payment method.
FR-06 Interactive filtering	Provide consistent date, store, bike-model and payment filters across pages.
FR-07 Validation and traceability	Cross-check measures in SQL/Excel; retain query, transformation and issue notes.

10.3 Non-functional requirements

Quality attribute	Requirement / acceptance check
Accuracy	Reconcile Power BI sales totals with SQL/Excel; document unresolved differences.
Usability	Use consistent colors, typography, labels, number formats and clear slicers across four pages.
Performance	Standard slicer interactions should usually respond within 5 seconds on the agreed test machine; record exceptions.
Security and privacy	Never commit secrets. Aggregate customer information and restrict repository access to the team.
Maintainability	Document transformations, relationships, KPI definitions and assumptions; use clear filenames and regular commits.

10.4 Data requirements and constraints

The supplied model contains FactSales (100,000 rows) and six dimensions: DimDate, DimBike, DimStore, DimPayment, DimSalesperson and DimCustomer. Validate keys, relationships, data types and the FactSales baseline before analysis. DimCustomer contains repeated Customer_ID values across profile variants, so distinct customer counts must use Customer_ID and the handling of demographic variants must be documented. DimSalesperson supplies Salesperson_ID only, with no salesperson names.

Out of scope unless the source is extended: profit/cost/margin, discounts, inventory, returns, customer location, product category, competitor market share and sentiment. These are not confirmed in the supplied fields and must not be inferred from sales totals.